

sson et al., 2021). To gain early insights into potential productivity gains from genAI, this section documents worker-reported time savings and shows how this data can inform estimates of aggregate and industry/firm-level productivity gains.

## 5.1 How Much Do Workers Use Generative AI?

We begin by documenting sizable variation in the intensity of genAI use among workers. This is important because, as we show in the next section, workers who use genAI more intensively tend to experience greater time savings.

Figure 8a reports how much time genAI users spent using the technology in a given day. Specifically, we ask “*Please think back to the days LAST WEEK on which you used Generative AI at work. On average, how much time did you spend actively using Generative AI at work?*” Respondents could select from three options: 15 minutes or less per day, between 15 minutes and one hour per day, or more than an hour per day.<sup>13</sup>

Figure 8a shows that 36.9% of genAI users report using genAI for an hour or more per day at work, 42.8% used it for between 15 and 60 minutes per day, and 20.3% used it for less than 15 minutes per day. We also show these estimates separately by workers’ daily frequency of use. Frequency and intensity of genAI use are positively correlated. 65.7% of daily genAI users report using it an hour or more each day, compared to only 6.4% of those who used it in the last month but not the last week. Overall, our results indicate wide variation in the frequency and intensity of genAI use at work.

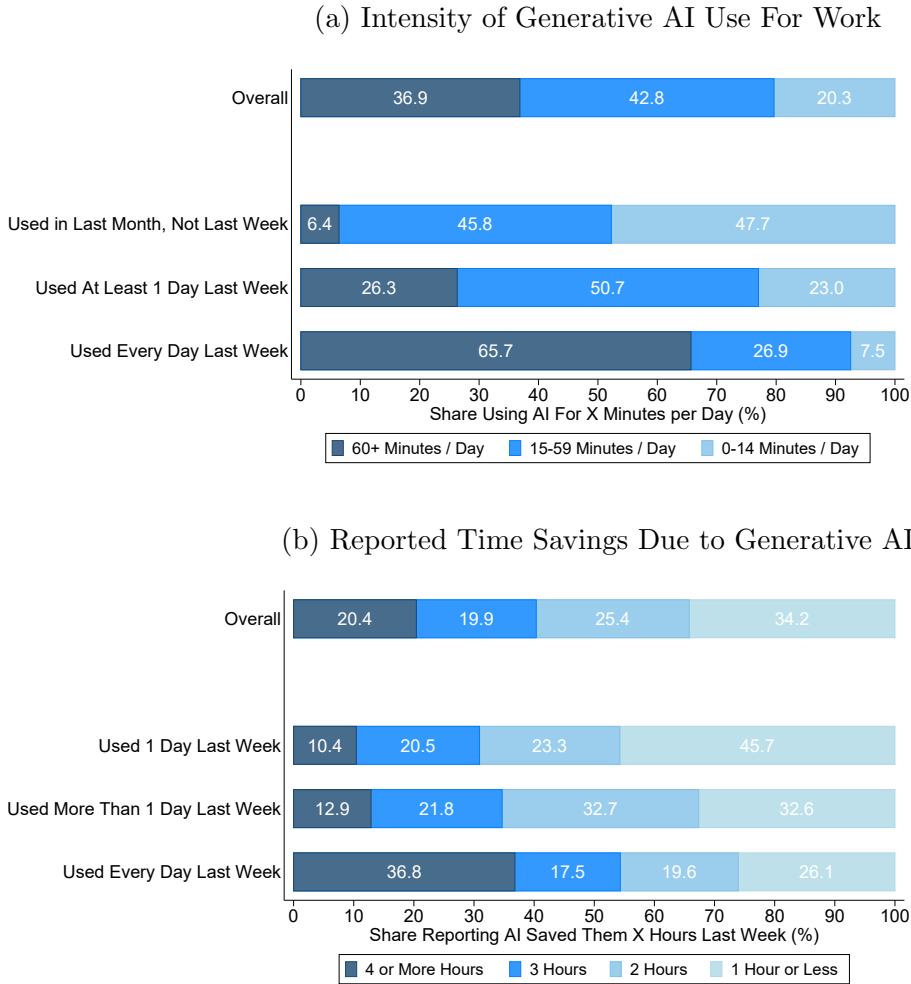
We estimate bounds on the share of total work hours assisted by genAI by combining data on usage intensity with data on days and hours worked in the previous week. (For this calculation, we restrict attention to our November 2024 wave, which contains more detailed intensity categories; see Appendix Figure B.2). For example, consider a worker who reports (i) working 40 hours last week over 5 days, (ii) using genAI on multiple but not all workdays last week, and (iii) using genAI for between 15 and 59 minutes per day on days that they used it. The lower bound implies the respondent used genAI on two days for 15 minutes each day ( $15/59 \cdot 2/40 = 1.3\%$  of their weekly hours); the upper bound implies using genAI on four workdays for 1 hour each day ( $1 \cdot 4/40 = 10\%$  of their weekly hours).

We first aggregate these estimates across the 26.4% of workers in the November 2024 wave who report using genAI in the past week. Among workers who use genAI, between 4.9 and 19.2% of all work hours were assisted by genAI. Next, we include the 73.6% of non-genAI-workers in our calculation, who spend no time using genAI by construction. Among all workers, between

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<sup>13</sup>Respondents who did not use genAI last week but did use it in the last four weeks were asked about their average usage on the days they used genAI over the last four weeks.

Figure 8: GenAI Intensity and Reported Time Savings



*Notes:* Panel (a) shows the distribution of daily time spent actively using genAI for work, among genAI users. The “Overall” bar reflects the distribution among all workers who use genAI. Other bars break down daily time use by daily frequency of use. Data source is the November 2024 waves of the RPS, ages 18-64. The sample is employed respondents who use genAI for work ( $N = 1000$ ). Panel (b) shows the distribution of work time saved in the previous week due to genAI, among individuals who use genAI for work. The “Overall” bar reflects the distribution among all workers who use genAI. Other bars break down time savings by daily frequency of use. Data source is the November 2024 wave of the RPS, ages 18-64 (the August 2024 wave did not ask about time savings). The sample is employed respondents who used genAI for work in the last week ( $N = 925$ ).

1.6% and 6.6% of total work hours were assisted by genAI.

## 5.2 Reported Time Savings Due to Generative AI

In the November 2024 wave, we asked workers who used genAI in the previous week to estimate how much time genAI saved them in the past week:

*You indicated that LAST WEEK you worked   X   hours and that you used Generative AI for your job. Now, imagine that LAST WEEK you did not have access to Generative AI. How many additional hours of work would you have needed to complete the same amount of work? [For X we fill in the respondent’s reported hours worked last week.]*

The answer options were: Less than 1 hour, 1 hour, 2 hours, 3 hours, 4 hours, More than 4 hours. For respondents who selected “less than 1 hour”, we assume that genAI saved zero hours of work. For respondents who selected “more than 4 hours,” we assume that genAI saved 4 hours of work. These assumptions are conservative in the sense that they reflect a lower bound of time savings.

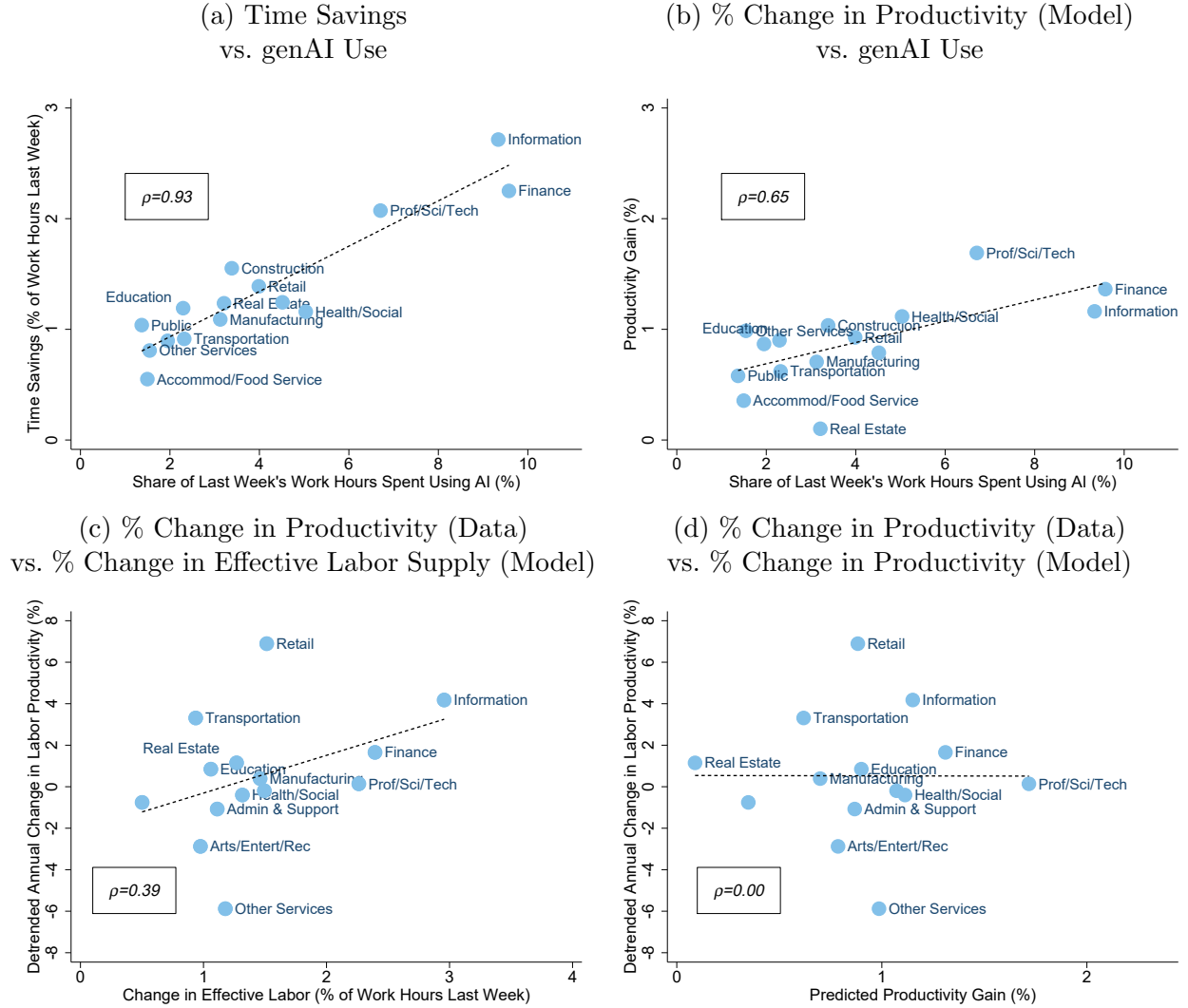
Figure 9 reports the results: 34.2% said genAI saved them an hour or less last week, 25.4% reported two hours, 19.9% reported three hours, and 20.4% reported four hours or more. More frequent users report higher average time savings. Among workers who used genAI every day last week, 36.8% reported saving four hours or more, compared to 10.4% of respondents who used it one day last week.

For each genAI user, we compute the share of working hours saved as the ratio of time saved last week to hours worked last week. We find an average time savings of 5.2% of work hours by genAI users in the November 2024 wave, or about 2.1 hours for a 40-weekly-hour worker. When we include all workers, including non-users, on average genAI saved workers saved 1.4% of total hours.

## 5.3 Potential Productivity Gains from Generative AI By Industry

We now present a standard production model to demonstrate how our data on genAI use and time savings can be used to estimate productivity gains on the industry or aggregate level. Detailed derivations can be found in Appendix D.1.

Figure 9: Generative AI Use, Time Savings, and Productivity Gains by Industry



*Notes:* Data source are the November 2024 wave of the RPS, ages 18-64 (the August 2024 wave did not ask about time savings). The sample is employed individuals (RPS,  $N = 3473$ ). Labor productivity is calculated as real value added (Bureau of Economic Analysis) divided by employment (Bureau of Labor Statistics) in each sector. We exclude Agriculture because the productivity cannot be constructed with the standard BLS data, and Mining, Utilities, Wholesale Trade, and Management industries due to low sample size. Panel (a) plots industry-wide time savings from genAI against industry-wide hours worked last week using genAI, with both variables expressed relative to total industry hours worked last week. Panel (b) plots estimated productivity gains, as defined in Equation (4), against the industry-wide hours using genAI last week relative to total industry hours worked last week. When calculating the mean share of last week's work hours spent using genAI, we weight observations by their last week's work hours to obtain an aggregate estimate of genAI usage as a share of hours worked. See text for details on the calculation of estimated productivity gains. Panels (c) and (d) plot estimated changes in effective labor and productivity against annualized labor productivity growth in the US over the years 2022 - 2024, detrended by subtracting the annualized growth predicted by the trend over the period 2015-2019 from the actual growth rate. To minimize the impact of outliers, panels (b), (c), and (d), which weight time savings using wages, exclude observations with (i) hourly wages below the national minimum wage of \$7.25 or (ii) weekly earnings above \$10,000.